

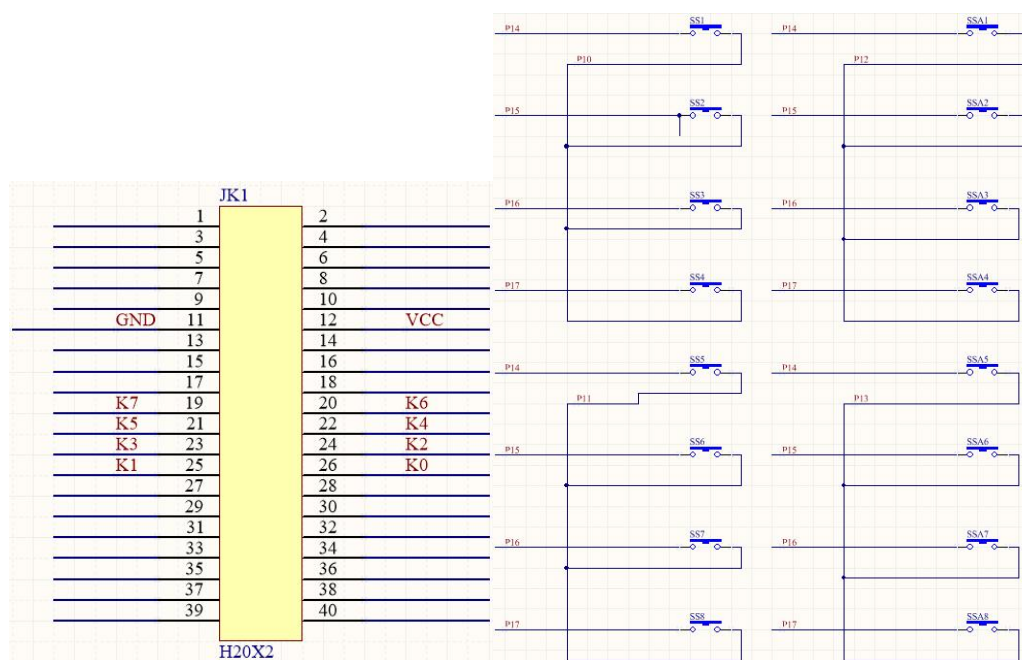
4X4 矩阵键盘扫描实验

1、实验目的：

了解 4X4 键盘的 8 根线扫描 16 个键的方法。

2、实验原理：

4X4 阵列键盘十分常用，图 4-41 是其电路和接口。假设其两个 4 位口，A[3:0]和 B[3:0]都有上拉电阻图 4-42。在应用中，当按下某键后，为了辨别和读取键信息，一种比较常用的方法是，向 A 口扫描输入一组分别只含一个 0 的 4 位数据，如 1110，1101，1011 等。若有键按下，则 B 口一定会输出对应的数据，这时，只要结合 A，B 口的数据，就能判断出键的位置。如当键 S0 按下，对于输入的 A=1110 时，那么输出的 B=0111。于是 {B,A}=0111_1110 就成了 S0 的代码。当检测到某个键时，对应输出键盘码。



矩阵键盘 40 芯接口原理图

矩阵键盘原理图图

3、实验程序：

```
module K4X4 (CLK, A, B, R);
    input CLK;
    input [3:0] A;
```

```

output [3:0] B;
output [3:0] R;
    reg [1:0] C ;          reg [3:0] R,B ;
always @ (posedge CLK) begin
    C<=C+1;
    case(C)
    0: B=4'B0111; 1: B=4'B1011; 2: B=4'B1101; 3: B=4'B1110;
    endcase
    case({B,A} )
    8'B0111_1110 : R=4'H0;   8'B0111_1101 : R=4'H1;
    8'B0111_1011 : R=4'H2;   8'B0111_0111 : R=4'H3;
    8'B1110_1110 : R=4'H4;   8'B1110_1101 : R=4'H5;
    8'B1110_1011 : R=4'H6;   8'B1110_0111 : R=4'H7;
    8'B1101_1110 : R=4'H8;   8'B1101_1101 : R=4'H9;
    8'B1101_1011 : R=4'HA;   8'B1101_0111 : R=4'HB;
    8'B1011_1110 : R=4'HC;   8'B1011_1101 : R=4'HD;
    8'B1011_1011 : R=4'HE;   8'B1011_0111 : R=4'HF;
    endcase    end    endmodule

```

4 、实验仿真:

根据描述语言进行仿真测试，分析波形。

5 、管脚约束:

```

set_pin_assignment { HA[0] } { LOCATION = H16; IOSTANDARD = LVCMOS33;
PULLTYPE = PULLUP; }

set_pin_assignment { HA[1] } { LOCATION = H15; IOSTANDARD = LVCMOS33;
PULLTYPE = PULLUP; }

set_pin_assignment { HA[2] } { LOCATION = J16; IOSTANDARD = LVCMOS33;
PULLTYPE = PULLUP; }

set_pin_assignment { HA[3] } { LOCATION = K15; IOSTANDARD = LVCMOS33;
PULLTYPE = PULLUP; }

set_pin_assignment { LB[0] } { LOCATION = G14; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }

set_pin_assignment { LB[1] } { LOCATION = G16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }

```

```
set_pin_assignment { LB[2] } { LOCATION = F15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { LB[3] } { LOCATION = F14; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { SCLK } { LOCATION = R7; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }  
  
set_pin_assignment { dataout[0] } { LOCATION = C15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[1] } { LOCATION = C16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[2] } { LOCATION = B6; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[3] } { LOCATION = A5; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[4] } { LOCATION = B5; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[5] } { LOCATION = A8; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { dataout[6] } { LOCATION = A7; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[0] } { LOCATION = D11; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[1] } { LOCATION = E11; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[2] } { LOCATION = F10; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[3] } { LOCATION = C13; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[4] } { LOCATION = E10; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[5] } { LOCATION = E12; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
```

```
set_pin_assignment { en[6] } { LOCATION = E16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
  
set_pin_assignment { en[7] } { LOCATION = F16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
```

6 、下载测试

将 4X4 键盘模块插在开发板 J1 口上。

下载程序之后。点击任意键，会看到数码管显示译码数；

